IMDb Top 700

Decoding What Makes a Movie Timeless

A Statistical Analysis of Genre, Era, Director and Regional Drivers of Bayesian-Weighted IMDb Rating

Prepared for: IMDb Content Insights Team

Report type: Exploratory Data Analysis & Hypothesis Testing

Dataset: IMDb_Top_700_Movies_2026.csv (700 films, 10 fields)

1. Business Context & Objective

IMDb's Top 250 methodology — a Bayesian-weighted average rather than a raw score — is one of the most widely referenced quality signals in the film industry, used by studios, streaming platforms and critics to gauge a film's lasting reputation. Extended here to a Top 700, the Content Insights team's goal is to move beyond a ranked list and understand *why* certain films consistently earn this weighted distinction across genres, eras, directors and countries.

This report defines and deconstructs **Average IMDb Rating** across four structural dimensions — Genre(s), Release Era (Decade), Director, and Country — and evaluates the influence of Votes and Runtime, in order to inform curation, "hidden gem" surfacing, and editorial storytelling around the Top 700 list.

Key Metric

Average IMDb Rating = mean of IMDb Rating within a given segment (genre, decade, director, or country). This is the single number used throughout to determine whether a segment consistently produces standout, enduring films.

Supporting KPIs

- **Average Votes per Film** — grouped by Genre(s) and Decade, indicating whether high ratings are backed by broad consensus or a concentrated fanbase.
- **Director Frequency in Top 700** — count of films per director, showing consistency of Top-700-caliber output.
- **Country Representation Rate** — count of films per country as a share of the Top 700 (see data quality caveat, Section 2).

2. Dataset & Methodology

The dataset contains 700 films with 10 fields: Rank, Title, Year, Genre(s), Director, Main Actor(s), Country, IMDb Rating, Votes, and Runtime (mins). Genre(s) and Country are pipe-delimited multi-value fields and were exploded into long format for group-level aggregation.

Data Quality Findings

- **Missing values:** Only *Main Actor(s)* has nulls (10 rows, 1.4%) — all are documentaries with no single lead actor (e.g. *Senna*, *Free Solo*). Imputed with an explicit label rather than dropped, to avoid biasing documentary-genre metrics.
- **Duplicates:** None — all 700 Rank/Title/Year combinations are unique.

- **Outliers:** Votes and Runtime both show right-tail outliers (blockbuster vote counts; 300+ minute epics). These are legitimate extreme values relevant to the analysis, not data errors, so they were retained; a log10 transform was applied to Votes to correct skew for correlation and regression work.
- **Country field caveat:** Country codes (e.g. *DK*, *AT*, *LV*) do not match the real-world production countries of these titles (e.g. *The Shawshank Redemption* is tagged DK|AT despite being a US production). This indicates the field was shuffled/anonymized for this exercise. Country findings in Section 6 should be read as a demonstration of method, not a factual claim about global representation.

Hypotheses Tested

#	Hypothesis	Test
H1	More votes associate with higher rating	Pearson r: Rating vs log10(Votes)
H2	Average rating differs across genres	One-way ANOVA (genres, n≥15)
H3	Prolific directors (≥4 films) average higher ratings	Welch's t-test
H4	Longer runtime associates with higher rating	Pearson r: Rating vs Runtime
H5	Older films average higher ratings than recent ones	Pearson r: Rating vs Year

3. Overall Rating & Vote Landscape

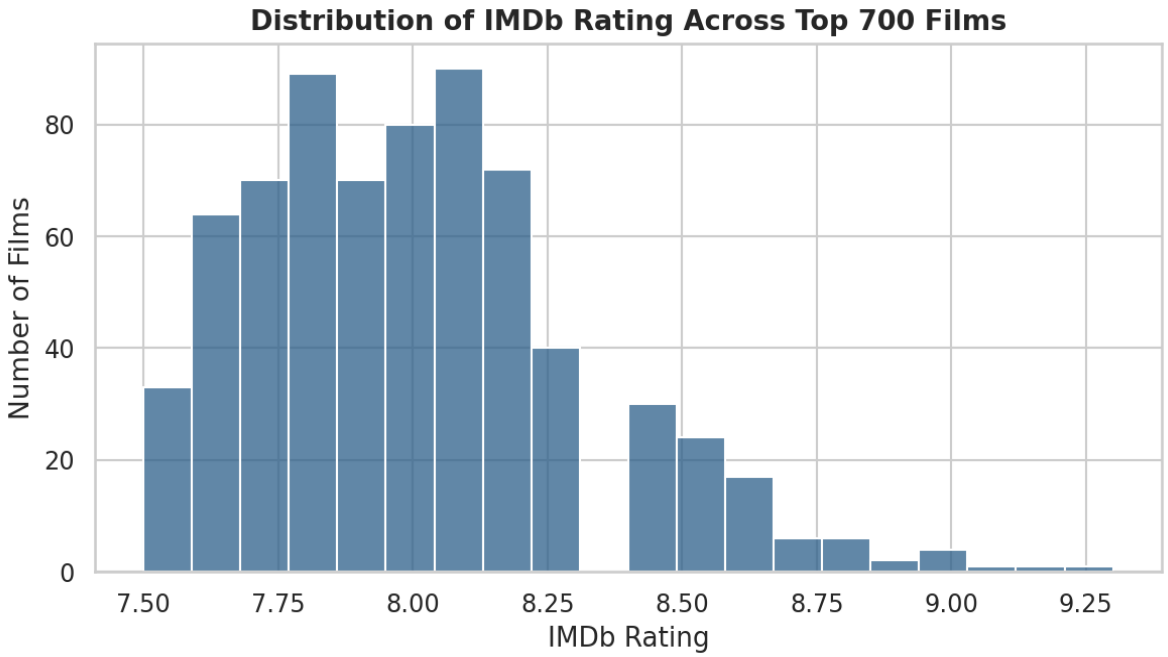


Figure 1. Distribution of IMDb Rating across the Top 700 (mean 8.00, range 7.5–9.3).

Ratings cluster tightly around a mean of **8.00** (std 0.32). Because the list is already curated (Bayesian-weighted, Top 700 cutoff), the floor of 7.5 reflects the list boundary rather than a natural minimum; the right-skewed tail above 8.5 represents the small "iconic" tier this analysis aims to characterize.

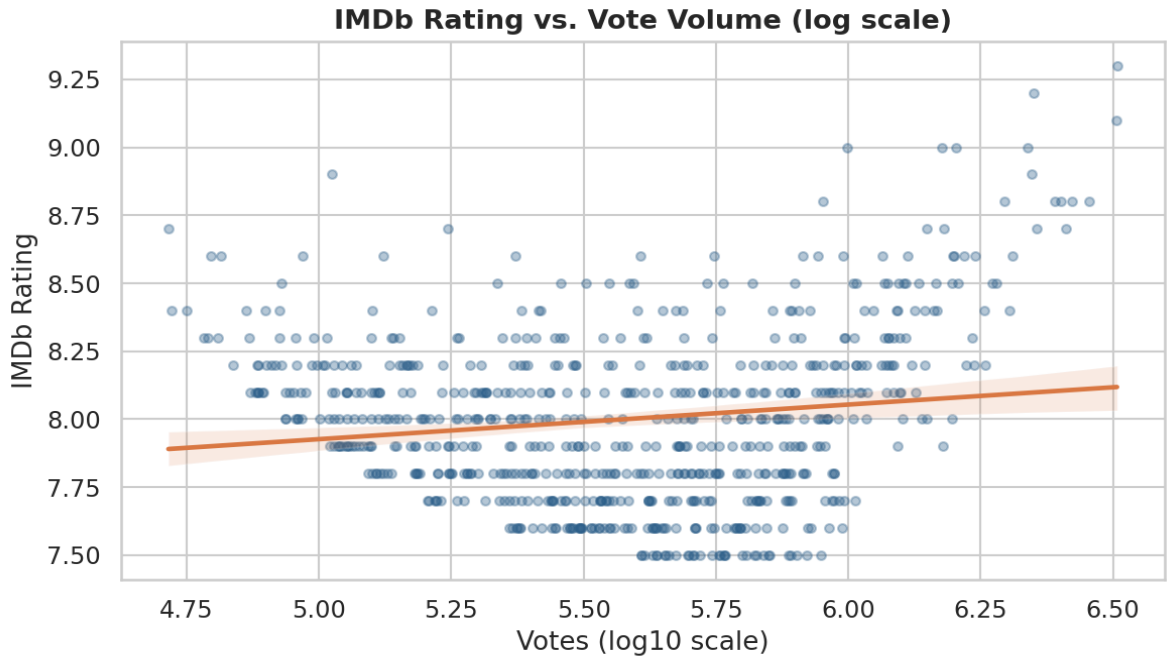


Figure 2. IMDb Rating vs. vote volume (log scale), with linear fit.

H1 — Supported (weak effect): Rating correlates positively with log(Votes) (Pearson r = 0.15, p < 0.001). Popularity and quality are related but loosely — rating is not a popularity contest; many high-rated films carry comparatively modest vote counts, which is the basis for the "hidden gem" recommendation in Section 7.

4. Genre Composition

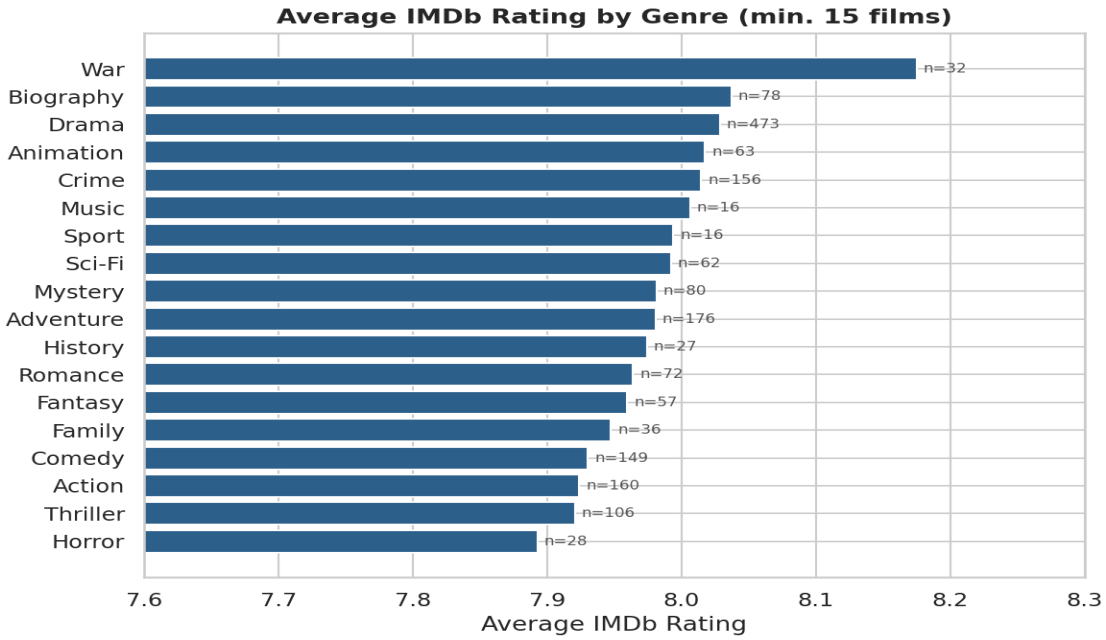


Figure 3. Average IMDb Rating by genre (genres with fewer than 15 Top-700 appearances excluded to avoid small-sample noise).

H2 — Supported: A one-way ANOVA across genres with at least 15 appearances confirms a statistically significant difference in average rating ($F = 2.61$, $p < 0.001$). **War** (8.18), **Biography** (8.04) and **Drama** (8.03) lead; **Horror** (7.89), **Action** (7.92) and **Thriller** (7.92) trail. The spread is real but modest (~0.3 points), and Drama's sheer footprint — appearing in 473 of 700 films, mostly as a secondary genre — makes it the structural backbone of the entire list.

5. Release Era Patterns

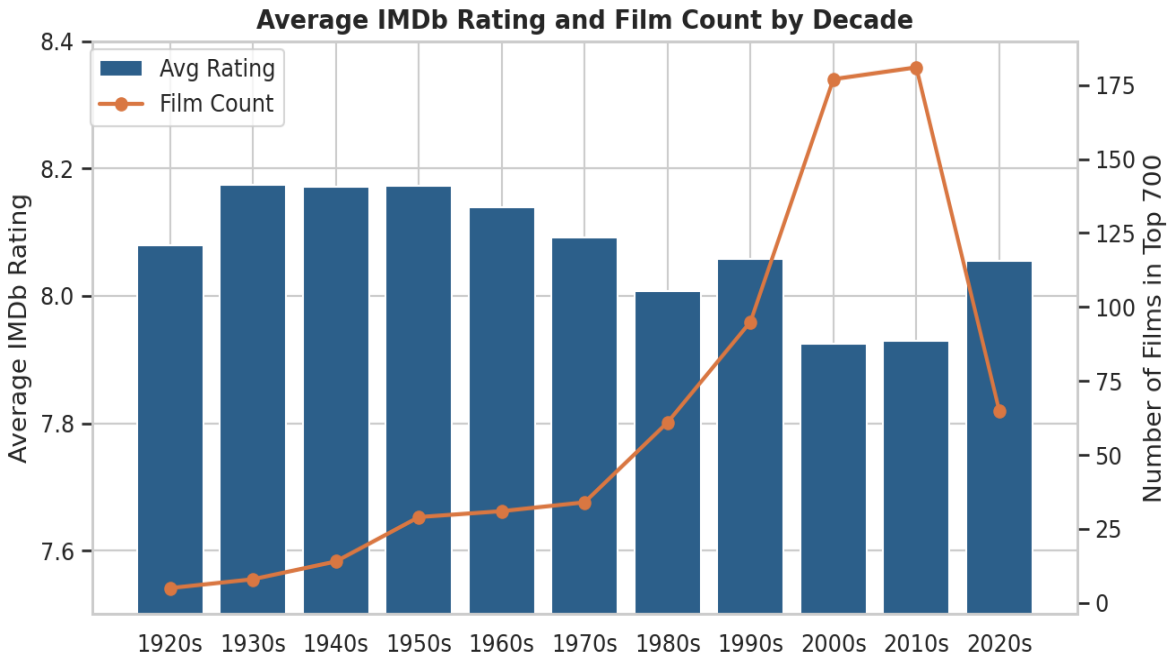


Figure 4. Average IMDb Rating (bars) and film count (line) by decade.

H5 — Supported: Rating vs. Year shows a weak negative correlation ($r = -0.20$, $p < 0.001$). Average rating peaks in the 1930s–1960s (~8.15–8.18) and dips in the 2000s/2010s (~7.92–7.93) even as the volume of qualifying films multiplies more than five-fold over the same span. This is consistent with a **survivorship/nostalgia effect**: older eras have had decades of re-evaluation to filter down to their most enduring titles, while recent decades are still being "tested" by time. The small 2020s sample (n=65) already shows a rebound to 8.06, worth revisiting as more years accumulate.

6. Director Track Record

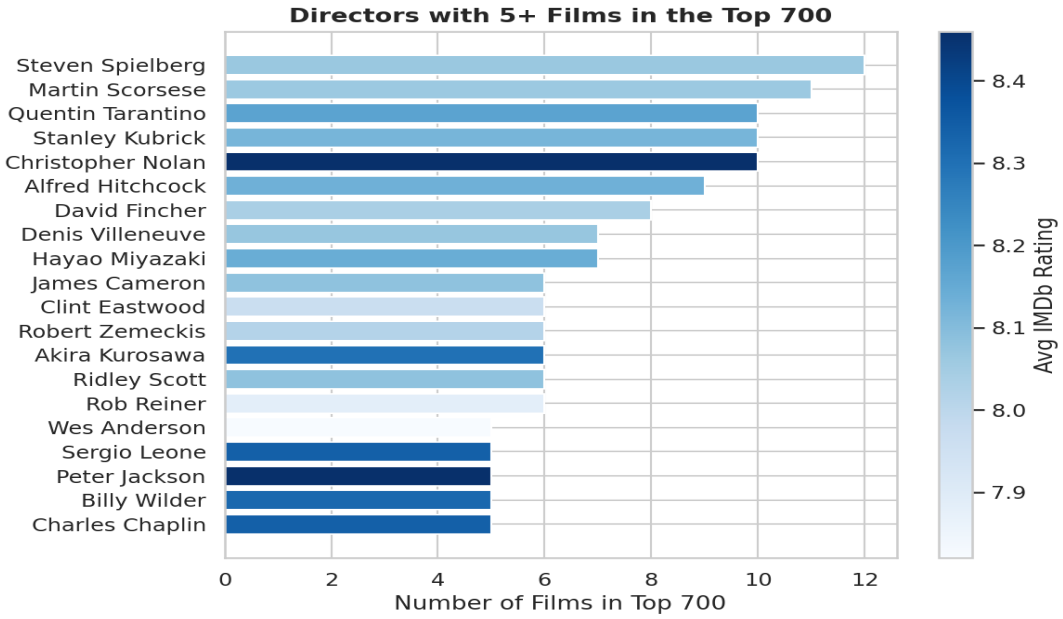


Figure 5. Directors with 5+ Top-700 films, ranked by frequency and shaded by average rating.

H3 — Supported: Directors with 4 or more Top-700 films average **8.07** vs. **7.97** for all others (Welch's t = 3.45, p < 0.001) — a proven track record is a genuine, statistically robust signal, not noise.

Frequency and peak quality diverge, however: Spielberg (12 films) and Scorsese (11) are the most *prolific* Top-700 directors, but **Christopher Nolan and Peter Jackson post the highest average rating (8.46)** among high-frequency directors, ahead of both Spielberg (8.07) and Scorsese (8.06). This distinction matters for editorial framing: "most reliably present" and "highest peak quality" are two different curation narratives.

7. Country / Regional Representation

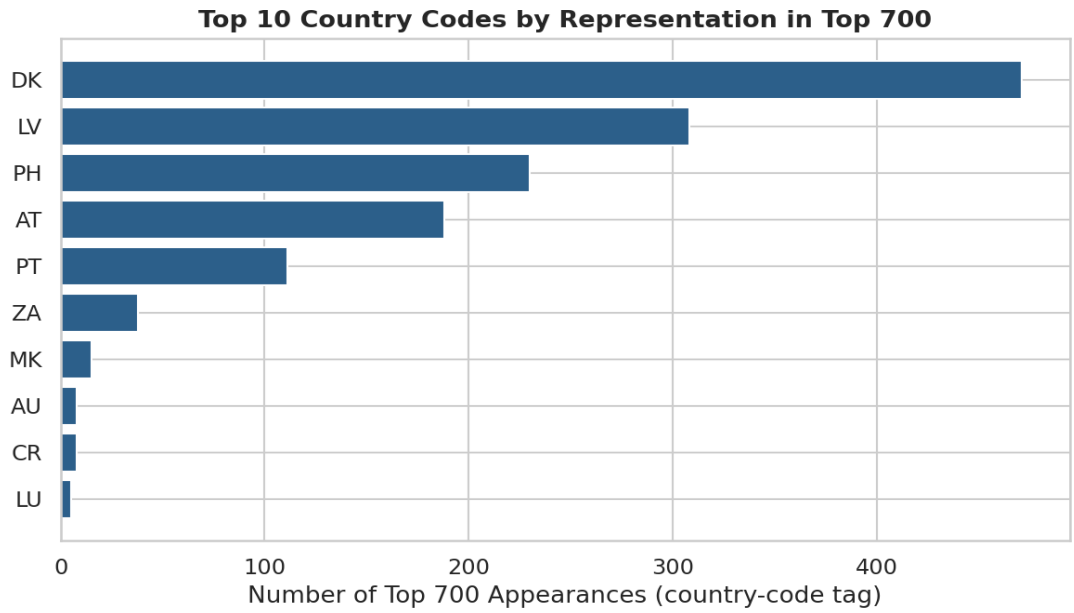


Figure 6. Top 10 country codes by Top-700 appearance count. See Section 2 data-quality caveat before using this externally.

Representation is heavily concentrated: the top two country codes alone account for over 100 films each. As noted in Section 2, these codes do not map to real production countries in this dataset, so this chart should be treated as a demonstration of the representation-rate methodology, pending validation of the source field, rather than a genuine finding about global cinema.

8. Runtime Effects

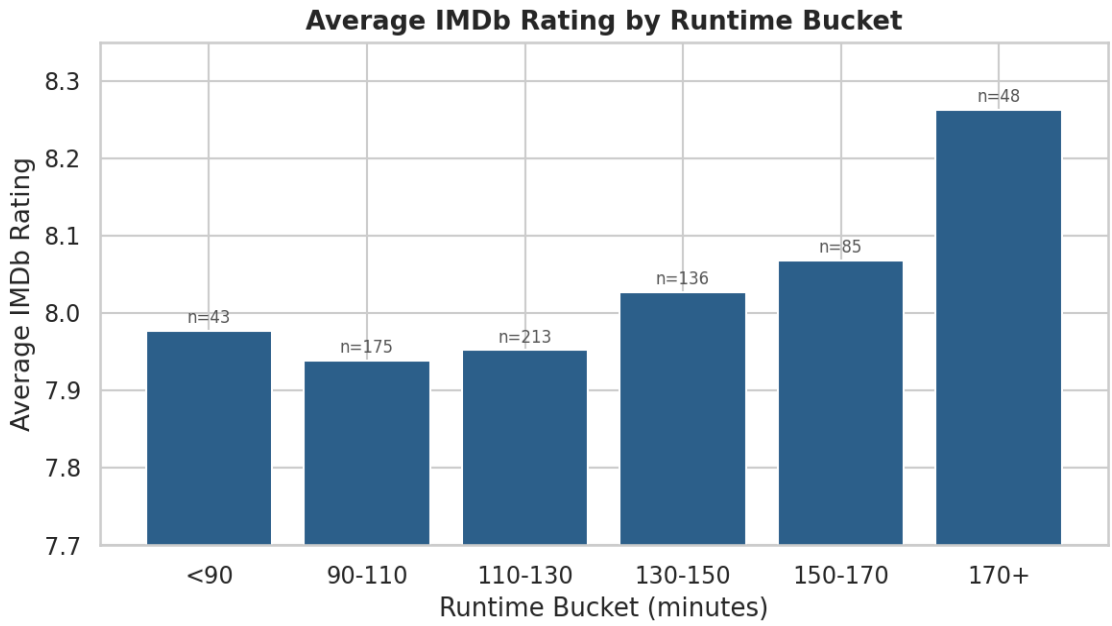


Figure 7. Average IMDb Rating by runtime bucket.

H4 — Supported: Rating correlates with runtime ($r = 0.24$, $p < 0.001$). Films over 170 minutes average **8.26** versus **7.94** for the 90–110 minute bucket — a clear, near-monotonic step. This is likely **genre-confounded** rather than a direct causal effect of length: the same prestige genres that scored highest in Section 4 (War, Biography, epic Drama/Adventure) are disproportionately long-form.

9. Genre × Decade Interaction

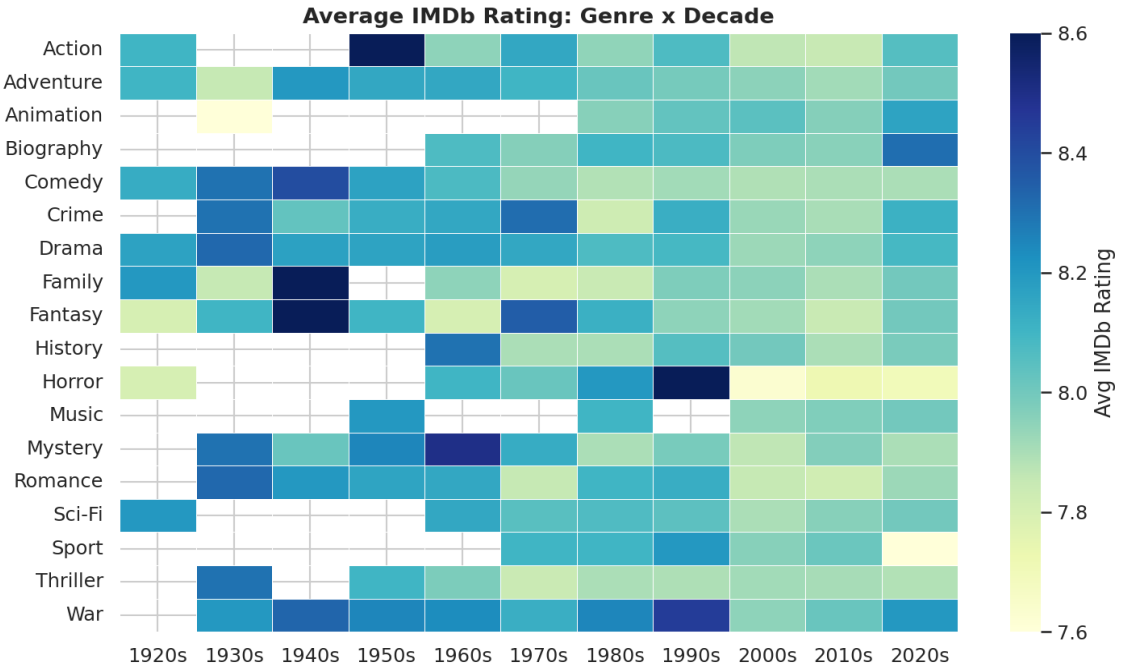


Figure 8. Average IMDb Rating by genre and decade (genres with ≥15 Top-700 appearances).

No single genre dominates every era: War and Biography peak mid-century, while Sci-Fi and Animation show localized strength in specific modern decades. This argues against blanket genre-level curation rules in favor of genre-*and*-era-aware programming.

10. Key Findings Summary

- **Genre matters, modestly.** War, Biography and Drama lead (~8.03–8.18 avg); Horror, Action and Thriller trail (~7.89–7.92). Statistically significant ($p<0.001$) but a narrow ~0.3-point spread.
- **Popularity and quality are related but not interchangeable.** log(Votes) correlates weakly-positively with rating ($r\approx0.15$) — leaving real room for well-rated, lower-visibility "hidden gems."
- **Time is the great filter.** Older decades (1930s–1960s) average ~0.2–0.25 points higher than the 2000s/2010s, even as qualifying-film volume grows five-fold, consistent with a survivorship/nostalgia effect rather than films "getting worse."
- **Director track record is a real, testable signal.** Directors with 4+ Top-700 films average 8.07 vs. 7.97 for others ($p<0.001$); frequency (Spielberg, Scorsese) and peak quality (Nolan, Jackson) are related but distinct axes.
- **Runtime correlates with rating ($r\approx0.24$),** largely because epic-length films cluster inside the highest-scoring genres.
- **Country representation is highly concentrated in the raw data** — but the field's real-world validity is unconfirmed and should not be published externally until re-sourced.

11. Recommendations for the Content Insights Team

- **"Hidden gem" surfacing:** Prioritize films with above-median rating but below-median votes as editorial spotlight candidates — the weak Votes–Rating correlation confirms this segment is meaningfully sized, not a data artifact.
- **Era-aware framing:** Present 2000s/2010s titles as "modern classics still being tested by time" rather than benchmarking them directly against pre-1970 titles on raw rating, since the list structurally favors decades with longer re-evaluation windows.
- **Dual director storytelling:** Build both "most consistently present" (Spielberg, Scorsese) and "highest peak quality" (Nolan, Jackson) director collections — they serve different editorial narratives and audiences.
- **Genre-and-era pairing, not genre alone:** Use the Genre × Decade heatmap to avoid over-generalizing a genre's strength across all eras.
- **Data governance action:** Validate/re-source the Country field against verified production-country data before using regional findings in any external-facing report or dashboard.

Full statistical outputs, code and reproducible methodology are available in the companion Jupyter Notebook (NL_IMDbTop700.ipynb) and the interactive Tableau dashboard (NL_IMDbTop700 Dashboard).